```
-- Person 3: T3.4 - T3.8 (PostgreSQL). Needs V3 (plans, plan_versions, product_va
CREATE SEQUENCE subscription_number_seq START 1;
CREATE SEQUENCE invoice_number_seq START 1;

CREATE TABLE quotation_templates (
    id            BIGSERIAL PRIMARY KEY,
    name          VARCHAR(150) NOT NULL,
    validity_days INT          NOT NULL CHECK (validity_days > 0),
    terms_text    TEXT
);

CREATE TABLE subscriptions (
    id                    BIGSERIAL PRIMARY KEY,
    number                VARCHAR(30) NOT NULL UNIQUE,
    customer_id           BIGINT      NOT NULL,            -- customers table bel
    plan_version_id       BIGINT      NOT NULL REFERENCES plan_versions(id),
    variant_id            BIGINT      REFERENCES product_variants(id),
    quantity              INT         NOT NULL DEFAULT 1 CHECK (quantity > 0),
    unit_price            BIGINT      NOT NULL,            -- paise, locked at cr
    status                VARCHAR(20) NOT NULL DEFAULT 'DRAFT'
                          CHECK (status IN ('DRAFT','QUOTATION','CONFIRMED','ACTI
    start_date            DATE,
    expiry_date           DATE,
    next_billing_date     DATE,
    is_paused             BOOLEAN     NOT NULL DEFAULT FALSE,
    quotation_template_id BIGINT      REFERENCES quotation_templates(id),
    quotation_valid_until DATE,
    version               BIGINT      NOT NULL DEFAULT 0   -- optimistic locking
);
CREATE INDEX idx_subscriptions_billing ON subscriptions (status, is_paused, next_
CREATE INDEX idx_subscriptions_customer ON subscriptions (customer_id);

CREATE TABLE tax_rules (
    id               BIGSERIAL PRIMARY KEY,
    name             VARCHAR(100) NOT NULL,
    rate_basis_points INT         NOT NULL CHECK (rate_basis_points BETWEEN 0 AN
    tax_type         VARCHAR(10)  NOT NULL CHECK (tax_type IN ('GST','VAT','SALE
    is_default       BOOLEAN      NOT NULL DEFAULT FALSE,
    active           BOOLEAN      NOT NULL DEFAULT TRUE
);
CREATE UNIQUE INDEX uq_tax_rules_single_default ON tax_rules (is_default) WHERE i

CREATE TABLE job_runs (
```

```
    id              BIGSERIAL PRIMARY KEY,
    job_type        VARCHAR(40) NOT NULL,
    started_at      TIMESTAMP   NOT NULL,
    finished_at     TIMESTAMP,
    status          VARCHAR(30) NOT NULL,
    processed_count INT         NOT NULL DEFAULT 0,
    success_count   INT         NOT NULL DEFAULT 0,
    failure_count   INT         NOT NULL DEFAULT 0
);

CREATE TABLE invoices (
    id              BIGSERIAL PRIMARY KEY,
    number          VARCHAR(30) NOT NULL UNIQUE,
    subscription_id BIGINT      NOT NULL REFERENCES subscriptions(id),
    customer_id     BIGINT      NOT NULL,
    status          VARCHAR(20) NOT NULL DEFAULT 'DRAFT'
                    CHECK (status IN ('DRAFT','CONFIRMED','PAID','CANCELLED')),
    subtotal        BIGINT      NOT NULL,
    tax_total       BIGINT      NOT NULL,
    discount_total  BIGINT      NOT NULL,
    total_amount    BIGINT      NOT NULL,
    period_start    DATE        NOT NULL,
    period_end      DATE        NOT NULL,
    due_date        DATE,
    created_at      TIMESTAMP   NOT NULL DEFAULT now(),
    version         BIGINT      NOT NULL DEFAULT 0,
    CONSTRAINT uq_invoice_subscription_period UNIQUE (subscription_id, period_sta
);

CREATE TABLE invoice_lines (
    id               BIGSERIAL PRIMARY KEY,
    invoice_id       BIGINT       NOT NULL REFERENCES invoices(id),
    description      VARCHAR(255) NOT NULL,
    quantity         INT          NOT NULL,
    unit_price       BIGINT       NOT NULL,
    line_subtotal    BIGINT       NOT NULL,
    discount_amount  BIGINT       NOT NULL DEFAULT 0,
    tax_rule_id      BIGINT,
    tax_basis_points INT          NOT NULL DEFAULT 0,
    tax_amount       BIGINT       NOT NULL DEFAULT 0,
    line_total       BIGINT       NOT NULL
);
```